

TEST DESIGN DOCUMENT

Fake News Detection Using Social Media Text

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1.Introduction

The purpose of this Test Design Document is to outline the complete testing framework for the Fake News Detection System developed using NLP and transformer-based models. This document defines how the system will be tested to ensure correct classification of textual data as REAL or FAKE. It provides clarity on the test strategy, coverage, scope, and conditions under which the model should operate successfully. The testing methodology ensures that both the backend model and the Streamlit user interface behave consistently across a variety of input scenarios. It also helps identify and address potential weaknesses in prediction accuracy and user interaction.

2.Objective Of Test Design

The main objective of this test design is to clearly define the testing conditions required to validate the performance and stability of the Fake News Detection System. This includes creating structured test cases, defining expected outcomes, and documenting how different types of text inputs will be interpreted by the model. The test design ensures that functional, usability, and edge-case testing conditions are met before final deployment. It aims to detect any incorrect classifications, UI inconsistencies, or failures in system behavior. Overall, the objective is to guarantee that the system operates reliably under real-world usage.

3.Test Coverage

The test coverage includes all core functionalities related to the prediction engine and user interface. This involves testing the model's ability to classify real news, fake news, ambiguous inputs, exaggerations, and fictional statements. Coverage includes validating the accuracy and consistency of confidence scores for each prediction. UI behavior will also be evaluated, including handling empty inputs, error messages, and responsiveness.

4.Test Data Requirements

The testing process requires a diversified set of inputs to evaluate the robustness of the model. This includes real, verified news headlines from reliable sources, as well as intentionally fake or fictional statements. Ambiguous social media posts, sarcastic statements, and exaggerations should also be included to test edge-case prediction capability. Invalid input types, such as empty text or numerical-only inputs.

5.Test Approach

A black-box testing approach will be utilized, focusing on system outputs rather than internal model logic. All tests will be executed using the Streamlit UI, simulating a real user entering text for classification. Test cases will be executed manually, and predictions will be compared against expected outcomes. Test results will be recorded.

6.Entry Criteria

- The transformer model must be fully trained and exported correctly.
- Tokenizer, model folder, and Streamlit UI should be properly integrated.
- All required Python libraries and dependencies should be installed.
- The test environment should be functional and free of critical errors.
- The system should allow input submission and output display.

7. Exit Criteria

The testing process will be considered complete when all identified test cases are executed successfully according to the test plan. The system must consistently classify social media text inputs as either real or fake without system crashes or unexpected behavior. All critical and high-priority defects must be resolved or documented. The application should display prediction results correctly through the Streamlit interface for valid inputs and handle invalid or empty inputs gracefully.

8. Assumptions

- The NLP-based classification model is already trained and available for inference
- The system is executed in a local development environment
- Users provide news content in English language
- The frontend interface is properly connected to the backend logic
- Network connectivity is stable during testing

9. Risks

- The model may incorrectly classify ambiguous or misleading news content
- Dependency or environment-related issues may impact execution
- Performance may degrade when processing long or complex text inputs
- Changes in social media language trends may reduce prediction accuracy over time

10. Test Design Sheets

Test design sheets provide a structured representation of test cases used to validate the Fake News Detection System. The test cases are designed to verify core functionalities such as fake news detection, real news detection, and handling of invalid or empty inputs. Each test case defines the test steps, expected results, and actual outcomes. Detailed execution results are documented separately in the Test Case Document to ensure clarity and traceability.